

One Network, Per-Recording Physics: Shared-Weight Inertial Velocity Estimation for Cars, Quadrupeds, Humans and Drones

Technical Report for the IMU Odometry Challenge

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Abstract— We estimate one-second mean body-frame velocity from raw 200 Hz IMU data of cars, quadrupeds, handheld devices and drones with a single network and one shared set of weights: a 12-block, width-384 Conformer that reads the raw inertial signal and a gravity-aligned copy of it, carries eight learnable registers and three soft feed-forward expert mixtures, and predicts dense and one-second velocity, per-axis uncertainty and a platform posterior. Velocity regression is shared across platforms; nothing is dispatched to per-platform experts. A second, parameter-free stage reconciles the predictions with each recording’s own measurements: an uncertainty-weighted velocity, gravity and bias smoother with exact discrete mean factors, a frame-aware attitude and inertial fit for aerial recordings whose IMU is expressed in world coordinates, and drag and level calibrations for the body-frame aerial rig. At fixed weights this stage reduces the filtered-validation challenge score by 24.1%, from 0.164675 to 0.125051, improving every platform, whereas descriptor and latent readouts of the frozen network give no further velocity gain and seven extended training continuations end in a narrow error range, which we read as practical representation saturation within the tested models and data. On the official per-sequence scoring service the submitted predictions score 0.16015 over all 89 sequences (macro AVE 0.12428 m/s, macro ATE₂₀ 0.45793 m); the Kaggle public score is 0.20589. The remaining error is concentrated in two fast flights of an aerial rig unseen at those speeds in the labelled data.

I. INTRODUCTION

A single inertial odometry model for several robot bodies faces three coupled difficulties. Velocity is not a function of the inertial measurements alone (adding a constant world velocity leaves an ideal IMU unchanged), so it must come from motion priors that differ between a wheeled vehicle, a legged robot, a hand and a rotorcraft; learned inertial odometry supplies such priors by regressing velocity or displacement from windows of IMU data [7], [6], [1]. The recordings differ in ways the platform label does not describe: in the challenge data we find aerial recordings whose accelerometer and gyroscope are expressed in world coordinates, and body-frame aerial recordings whose accelerometer quantization differs by an order of magnitude. And the score couples a velocity error with a position error obtained by integrating the predicted velocities with ground-truth orientation, so slowly varying per-recording biases that a per-window loss barely sees dominate it.

We answer with one shared learned component. A Conformer trained on all platforms at once regresses velocity, calibrates a per-axis uncertainty and classifies the platform: one set of weights, the embodiment inferred inside. A parameter-free estimator then treats the network output as a noisy measurement of velocity and solves, per recording, for the velocity, body-frame gravity and sensor biases that best reconcile these measurements with the kinematic and gravity-transport equations of the IMU, weighting each measurement by the network’s own uncertainty, and is instantiated with different physics where a recording’s coordinate convention or sensor calls for it. The method is offline by design: the network is bidirectional within its tile and the estimator fits each recording as a whole (Sec. II-F). We make four points: (i) shared regression is viable and the soft experts do not become platform dispatch; (ii) physical and temporal estimation at fixed weights is worth more than every architectural change we tried, 24.1% of the filtered-validation score; (iii) the tested routes to a better network give diminishing returns, which we call practical representation saturation and bound carefully; (iv) the platform labels hide sensor and coordinate structure that is recoverable without labels and decides which physics applies.

II. METHOD

A. Overall Architecture

Fig. 1 shows the complete method. For a recording of T samples at 200 Hz, $N = T/20$ tokens and $W = N/10$ one-second windows, the shared predictor maps the $T \times 6$ inertial signal to a dense velocity $\hat{\mathbf{v}}_t \in \mathbb{R}^3$ per token, a window velocity $\hat{\mathbf{v}}_k$ and per-axis uncertainty $\hat{\sigma}_k \in \mathbb{R}_{>0}^3$ per window, and a four-class platform posterior per tile. The same weights are evaluated at 60, 120 and 240 windows of context with overlapping cosine-tapered tiles of stride 30. A numerical stage without learned parameters (Sec. II-E) consumes these outputs together with the raw accelerometer, gyroscope and timestamps.

Shared predictor. The twelve input channels are the accelerometer scaled by $g = 9.81 \text{ m/s}^2$ and the gyroscope scaled by 0.77 rad/s , followed by a gravity-aligned copy of both triads. A three-stage residual convolutional stem (widths 64/128/256, kernels 7/7/11, strides 2/2/5) reduces the 200 Hz signal to 10 Hz tokens of width 384. Eight learnable registers are prepended, and twelve pre-norm Conformer blocks [3]

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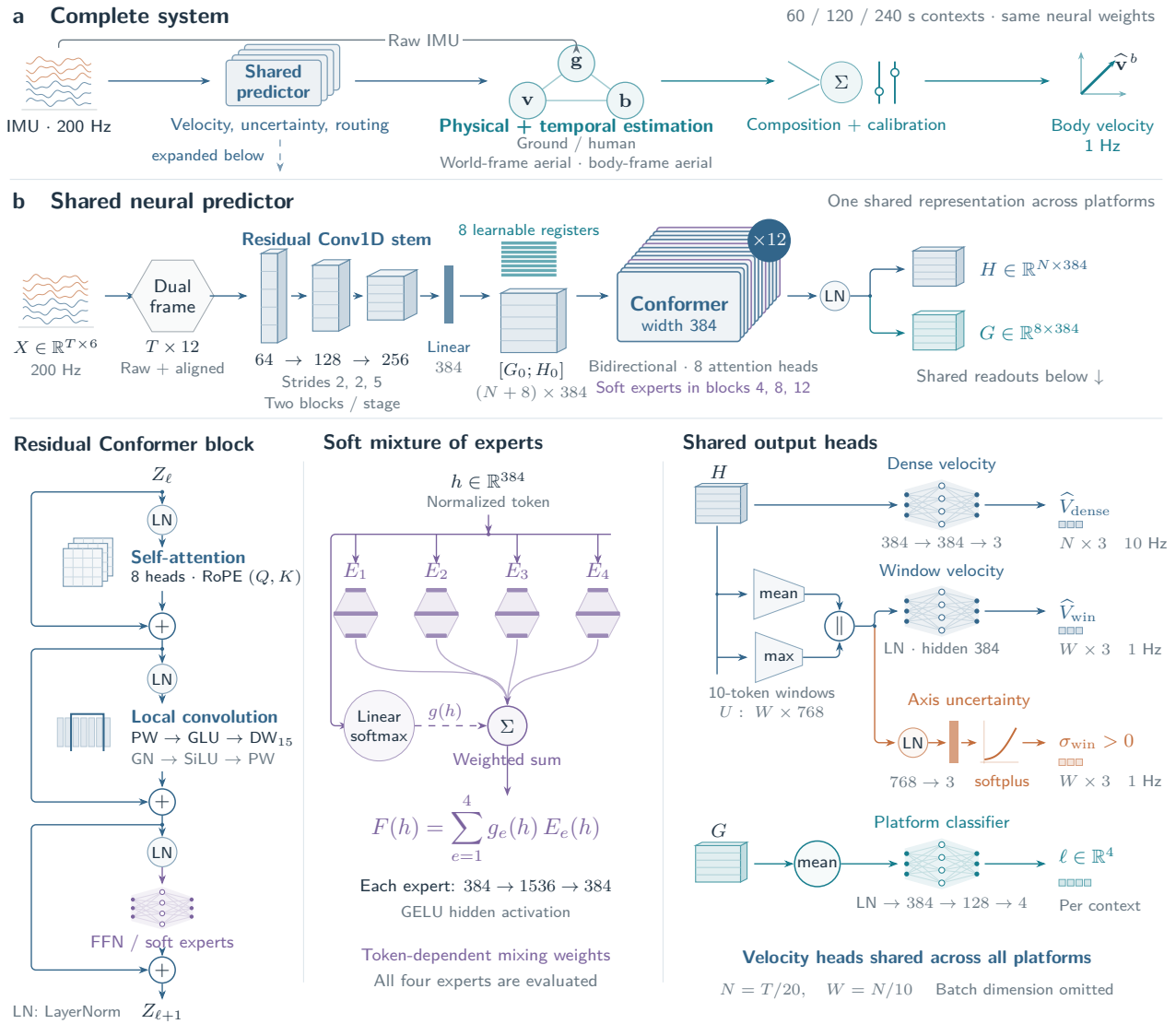


Fig. 1. Shared neural prediction with physical and temporal estimation. (a) One frozen network produces velocity, uncertainty and routing outputs at three context lengths from the same weights; recording-dependent numerical estimation combines them with the raw IMU, and composition and calibration produce one body-velocity vector per second. The estimator box summarises the ground/human and the two aerial families as a dependency overview, not an execution order. (b) The predictor: raw and gravity-aligned inputs pass a residual convolutional stem, eight learnable registers and twelve bidirectional Conformer blocks; blocks 4, 8 and 12 evaluate all four experts and mix them with token-dependent softmax weights, distinct from the register-based platform classifier. Ten data tokens are pooled by masked mean–max concatenation per window; velocity and uncertainty use separate heads.

apply bidirectional rotary attention (8 heads), a depthwise convolution module (kernel 15) and a feed-forward update of hidden width 1536. In blocks 4, 8 and 12 the feed-forward update is a per-token softmax mixture of four experts, $F(h) = \sum_{e=1}^4 g_e(h) E_e(h)$, $g = \text{softmax}(W_g h + b_g)$: every expert is evaluated for every token, a dense mixture inside a shared trunk rather than a dispatch. After a final LayerNorm the data tokens $H \in \mathbb{R}^{N \times 384}$ feed a dense velocity MLP, and the window representation $u_k \in \mathbb{R}^{768}$, the masked mean and max of H over the ten valid tokens of window k concatenated, feeds a layer-normalised velocity MLP and a separate uncertainty head (softplus, clipped to $[10^{-3}, 10]$). The mean of the eight register tokens feeds a small MLP with four platform logits. The network has 41.3 M parameters.

Design choices for cross-platform generalisation. The gravity-aligned copy removes the dependence on mounting orientation for the part of the signal that gravity explains while the raw copy keeps body-frame information; a label-free frame gate blends the aligned gyroscope back towards the raw one for recordings whose gyroscope noise floor or gravity-direction dynamics indicate a world-frame convention. Registers and the platform head let the network infer the embodiment; the numerical stage, not the network, carries the platform-dependent physics.

B. Data Processing

Inputs are the released 200 Hz accelerometer and gyroscope triads and timestamps, without resampling. The network works on tokens of 20 samples and windows of 200

samples. At inference a recording is tiled into contexts of 60, 120 or 240 windows with stride 30; outputs of overlapping tiles are averaged with a cosine taper floored at 10^{-3} , uncertainties as variances, and platform logits are turned into a posterior per tile and averaged over the recording; the dense velocity is blended one-to-one with the window head after pooling. The gravity direction used for alignment is the accelerometer low-passed over 401 samples (2 s); alignment is the shortest-arc rotation taking it to $-\hat{\mathbf{z}}$, per sample. No statistic of other recordings enters inference. Overlapping-tile averaging and multiple context sweeps constitute test-time augmentation (TTA), with the same network weights in every sweep.

C. Training Data

Only the official challenge data were used (Table I; test IMU enters inference only): no external datasets, synthetic data or pretrained weights. A recording-overlap audit identified nine validation recordings that share a physical recording with training recordings or carry a mispaired quadruped label; development measurements use the remaining 71 (the filtered validation set) unless the full set of 80 is stated. The training split was used as released, including two quadruped recordings whose IMU and pose tracks are mispaired.

TABLE I
DATASETS USED FOR MODEL DEVELOPMENT AND TRAINING.

Dataset	Type	Usage
Challenge train set	Official	Pretraining; velocity training; uncertainty and platform-head fitting
Challenge validation set	Official	Pretraining; uncertainty and platform-head fitting; checkpoint selection; calibration
External / synthetic data	—	None

D. Training and Optimization

Training proceeded in stages along one inherited checkpoint lineage. Unsupervised pretraining of the encoder (15,000 updates) used the IMU signals of the 475 training and validation recordings without labels, with a view-invariance objective adapted from LeJEPa [4] and its SIGReg regulariser. A formation stage (80,000 updates) trained the velocity heads with a distillation term from two earlier supervised networks that was switched off after 64,000 updates; four continuations of 184,000 selected updates followed (the third introduced the soft experts), then a final stage of pure continuation, roughly one million supervised updates in all. Every supervised velocity-training stage used the training split only (395 recordings), drew a platform uniformly with the aerial weight raised by 1.3 and a chunk of 30 to 60 consecutive windows, and minimised $L = \mathcal{H}_{\text{dense}} + \mathcal{H}_{\text{window}} + 0.3L_{\text{ATE}} + 0.01L_{\text{bal}}$: a Huber loss on the velocity error norm (knee 0.4 m/s for ground platforms, 0.15 m/s aerial) at the token and window rates, a trajectory-error surrogate integrating the predicted velocity in the world

frame with ground-truth orientation over 1 to 20 s horizons, and a load-balance term on the expert weights. Augmentation applied random rotations of up to 13° to signal and labels, time warping by factors 0.4 to 2.5 with regenerated labels, small scale and cross-axis errors, turn-on biases, bias random walks, white noise and occasional saturation. AdamW (weight decay 0.05, batch 8, gradient-norm clip 1.0, FP32) ran at 3×10^{-4} with warm-up and cosine decay for pretraining and formation and at a constant 10^{-5} for every continuation (ten times higher for the expert parameters). An exponential moving average of the weights (decay 0.999) was evaluated every 2,000 updates on all 80 validation recordings, and each stage handed its validation-minimum EMA checkpoint to the next. The uncertainty head (Gaussian negative log-likelihood) and the platform classifier (class-balanced cross-entropy, label smoothing 0.05) were fitted afterwards on the frozen trunk with training and validation recordings; neither enters the velocity regression. Stages ran with fixed seeds on single cloud GPUs (Table II); every reported evaluation is one deterministic inference.

E. Inference

The complete pipeline is the released `inference.py`. (1) The network runs four times with the same weights: at 120 windows on the accelerator for base estimation, at 120 windows on the CPU (five threads) for the refinement inputs and the raw direction hedge, at 60 windows for body-frame aerial refinement and at 240 for humans. (2) Base estimation routes each recording from its platform posterior and runs three cumulative configurations of the smoother: ground platforms only; plus the aerial smoother behind a label-free frame gate; plus a speed gate that leaves ground recordings slower than 0.33 m/s raw. (3) World-frame aerial recordings receive the attitude and inertial fit. (4) Car, quadruped and human recordings are re-solved with exact 200 Hz mean factors, humans at 240 windows, quadruped and human with an uncertainty ordered by the recording's own stationarity fraction. (5) Body-frame aerial recordings are re-solved from the 60-window context with drag and level calibration.

Smoother. For a recording with token velocities $\hat{\mathbf{v}}_t$, uncertainties σ_t , token-mean specific force $\mathbf{f}_t$, angular rate $\boldsymbol{\omega}_t$ and durations Δt_t , the state holds a body velocity $\mathbf{v}_t$ and body gravity $\mathbf{g}_t$ per token and shared accelerometer and gyroscope biases $\mathbf{b}_a, \mathbf{b}_g$, a rotation $\boldsymbol{\theta}$, scale s and offset $\mathbf{b}_v$ relating the network's frame to the label frame (plus a lever arm for ground platforms). Gauss-Newton (three iterations) minimises

$$\begin{aligned}
J = & \sum_t \rho \left(\left\| \frac{1}{s} (sR(\boldsymbol{\theta})\hat{\mathbf{v}}_t + \mathbf{b}_v - \mathbf{v}_t) \right\|_{\Sigma_t} \right) + \sum_t \frac{\|\mathbf{r}_t^{\text{kin}}\|^2}{\sigma_{\text{kin}}^2 \Delta t_t} \\
& + \sum_t \frac{\|\mathbf{g}_{t+1} - \exp(-(\boldsymbol{\omega}_t - \mathbf{b}_g)\Delta t_t)\mathbf{g}_t\|^2}{\sigma_{\text{grav}}^2 \Delta t_t} \\
& + \sum_t \frac{(\|\mathbf{g}_t\| - g)^2}{\sigma_{\text{gnorm}}^2} + \text{priors on } (\mathbf{b}_a, \mathbf{b}_g, \boldsymbol{\theta}, s, \mathbf{b}_v) \\
& + \text{zero-velocity terms,}
\end{aligned} \tag{1}$$

with $\mathbf{r}_t^{\text{kin}} = \mathbf{v}_{t+1} - \mathbf{v}_t - \Delta t_t(\mathbf{f}_t - \mathbf{b}_a + \mathbf{g}_t - (\boldsymbol{\omega}_t - \mathbf{b}_g) \times \mathbf{v}_t)$, ρ a Huber loss, Σ_t the network uncertainty raised to a

platform exponent and scaled, and zero-velocity pseudo-measurements on runs of tokens with near-gravity specific force, small rotation rate and small predicted speed. The kinematic and gravity factors are discretised at the token rate for base estimation and replaced by exact 200 Hz mean factors (midpoint integration of the raw samples between token centres) for the refinements. A fit whose rotation or scale leaves a fixed envelope is refitted with those states frozen; one that still fails returns the raw prediction. Platform settings were fitted out-of-fold or tuned on validation recordings.

World-frame aerial fit. For the aerial subgroup with world-frame measurements, $\mathbf{f}_m \approx \dot{\mathbf{v}}^w - \mathbf{g}^w$ and $D\boldsymbol{\omega}_m \approx \text{vee}(\dot{R}R^\top)$ with $D = \text{diag}(1, -1, -1)$, we integrate the gyroscope into relative rotations, integrate the specific force into a world velocity up to an initial velocity, accelerometer bias and (for recordings of at least 30 windows) bias drift, align the resulting body-frame window means with the network’s window velocities by Procrustes plus a gyroscope-bias search, and accept the fit when its normalised residual is below 0.50; fits with residual above 0.40 are blended more cautiously with the smoothed prediction.

Body-frame aerial refinement and calibration. Body-frame aerial recordings are re-solved with exact kinematic factors. When the fitted velocity scale exceeds one by more than 10% (no labelled recording deviates by more than 6.4%), a guarded refit runs and the speed of each window is replaced by a rotor-drag estimate, $v_k = \kappa \|\bar{\mathbf{f}}_{xy,k}^{\text{lp}} - \mathbf{c}\|$ [5], whose gain κ and quiet-sensor offset $\mathbf{c}$ were fitted on the labelled flights of the same rig; otherwise a scalar level calibration with the same drag coefficient is admitted for ratios in (1.1, 1.15]. Recordings with coarse accelerometer quantization use their own smoother settings and no calibration. Fixed vertical offsets and scale factors, each applied only inside a regime read from the recording’s own statistics (a persistent descent of the estimate along gravity; a moderate lateral specific force), complete the calibration.

F. Model Complexity and Runtime

An offline replay of the 89 test sequences (30,644 s of IMU) took 515.4 s end to end on an Apple M5 Pro laptop, 59 times faster than real time; the four network sweeps took 472.3 s and the remainder is loading, numerical estimation and CSV writing (Table II).

Deployment scope. Every stage is a smoother rather than a filter: the Conformer attends bidirectionally within a tile of 60 to 240 windows, each window’s output averages the tiles that overlap it, the platform posterior is averaged over the recording, and the estimator fits velocity, gravity, biases and calibration constants to the whole recording. The method therefore cannot serve as a causal, online odometry source. Two measurements bound the lag a delayed-output variant would need: raw predictions at 60, 120 and 240 windows differ by less than 0.001, and 64-second segments retained 95% of the whole-recording smoothing gain (Sec. III), so a version with about one minute of latency is plausible but unmeasured.

TABLE II
MODEL COMPLEXITY AND RUNTIME STATISTICS.

Property	Value
Number of parameters	41.3 M (float32)
Training hardware	1 × A100 / H100 (cloud)
Training time	71 min pretrain; 5–9 h per stage
Inference hardware	Apple M5 Pro (MPS + 5 CPU threads)
Inference speed	59× real time, end to end
120-window sweeps	59.8 s (MPS) / 241.0 s (CPU)
Sampled MPS peak	1.43 GiB (driver allocation)
Online capable	No (future context, whole recording)

III. DEVELOPMENT EXPERIMENTS

Numbers here are development scores from the released metric code on the filtered validation set (71 recordings) unless stated; one checkpoint’s evaluation noise is about 0.0004 and matched seed twins differ by up to 0.004.

Staged training approaches a narrow range. Fig. 4(a) (appendix) plots the full-validation score of both supervised lineages against cumulative updates along the inherited checkpoint lineage, and Fig. 4(b) enlarges the late range. Seven extended continuations of the rebuilt lineage’s trunk with different recipes (a learned label-free frame gate, gyro-aided gravity conditioning, aerial sampling raised two- and four-fold, compiled execution) all reached their 400,000-update budgets and ended between 0.1704 and 0.1723 against 0.1706 for the released checkpoint: five marginally below it, all seven above it on the filtered set. Budget completion is not convergence and the runs are not seed replicates, so we read diminishing returns rather than a common asymptote. Individual recipes moved the score by a few 10^{-4} (learned frame gate -0.0003 to -0.0005 at matched updates, gyro-aided gravity -0.0003 to -0.0006 , no additional gain from stacking them). Doubling and quadrupling the aerial sampling weight cut world-frame aerial AVE by 0.0031 and 0.0062 m/s while worsening coarse-quantization aerial (+0.0016) and ground (+0.0003 to +0.0005) errors, a trade-off across and within platforms.

What changed the network little. The soft experts entered the lineage with a -0.0023 full-validation gain confounded with 200,000 extra updates; two controlled introductions were worse (+0.0034, +0.0086) and the routing entropy stays near uniform (1.29–1.38 versus $\ln 4 = 1.386$), so the experts do not specialise by platform. Label-free pretraining changed the formation stage by +0.0017 against a masked-autoencoder initialisation, within the seed spread. Feeding handcrafted IMU descriptors to the trunk instead of routing on them was a null within seed noise, a frozen descriptor router inserted into a trained trunk was harmful (+0.017 to +0.028, all in body-frame aerial recordings), and a persistent memory across tiles was +0.0066 worse. Eight-fold rotation test-time augmentation changed the filtered validation by only 0.0004, within noise, and is not used; compiled execution changed the numerics enough to cost +0.0002 and was not adopted.

Numerical estimation at fixed weights. The direct ablation of the numerical stage on the released weights is Fig. 2

and Sec. IV-D: raw predictions 0.164675, core estimator 0.142535, complete released procedure 0.125051. On the earlier reference network, whole-recording fitting of the two core smoothers improved 0.1684 to 0.1553, and 64-second segments retained 95% of that gain.

Additional readouts of the frozen network do not help. We fitted eight prespecified residual readouts on the released weights with fold-local preprocessing (five grouped folds per platform; 71 recordings in 58 recording groups): ridge or boosted trees on the network outputs, optionally with 25 IMU descriptors and/or the 768-dimensional pooled representation. None improved platform-macro AVE (+0.0036 to +0.0416 m/s against 0.130265 m/s) and all eight worsened the local challenge score (0.171–0.219 against 0.1645); two boosted latent variants reduced quadruped AVE by 0.0008 and 0.0004 m/s with bootstrap intervals spanning zero. Group-held-out speed probes agree: the network’s speed output has lower MAE than a descriptor regressor fitted to training recordings on every platform (car 0.077 vs 0.171, quadruped 0.039 vs 0.140, human 0.039 vs 0.085, drone 0.114 vs 0.374 m/s), and adding descriptors to it does not beat the unchanged network.

IV. EXPERIMENTS

A. Official Challenge Performance

All numbers in Tables III, IV and V were produced by the official per-sequence scoring service, https://huggingface.co/spaces/Tartan-IMU/imu_odymetry_challenge_scoring, for the submitted CSV with md5 = **6d371534ca010cdc7fdc8f18d00fc6a0** (30,644 windows, 89 sequences). The same file scored 0.20589 on the Kaggle public leaderboard.

TABLE III

OVERALL PERFORMANCE FROM THE OFFICIAL PER-SEQUENCE SCORING SERVICE OVER ALL 89 TEST SEQUENCES. THE TARTANIMU SCORE IS 0.6 AVE/0.7356 + 0.4 ATE₂₀/3.1160, DIMENSIONLESS, LOWER IS BETTER; ATE₂₀ AND RTE IN M, AVE IN M/S. RTE (5 S) DOES NOT ENTER THE RANKING.

Method	Score ↓	ATE ₂₀ ↓	AVE ↓	RTE ↓
Ours	0.16015	0.45793	0.12428	0.62542

For context, the organisers’ evaluation page reports full-test AVE 0.461 m/s and ATE₂₀ 1.261 m for their unified baseline (an approximate full-test score of 0.538 from those rounded components; public-split score 0.637).¹

B. Per-Platform Results

The aerial platform contributes 56.6% of the score with 45 of the 89 sequences and only 2,099 of the 30,644 windows (Table IV): aerial AVE (0.330 m/s) is 4.5 to 9 times the ground values and aerial RTE 1.72 m against 0.21–0.34 m. The human platform has the lowest AVE (0.035 m/s) but

the largest ground ATE₂₀ (0.496 m), because the metric integrates small within-window rotation errors over long handheld recordings.

TABLE IV

PERFORMANCE BY PLATFORM FROM THE OFFICIAL SCORING SERVICE; THE AVERAGE WEIGHTS THE FOUR PLATFORMS EQUALLY.

Platform	ATE ₂₀ ↓	AVE ↓	RTE ↓	Seq.
Wheeled / Car	0.25997	0.05888	0.20782	18
Handheld / Human	0.49568	0.03497	0.23605	10
Legged / Quadruped	0.34896	0.07329	0.34242	16
Aerial / Drone	0.72712	0.32996	1.71540	45
Average (equal platform weight)	0.45793	0.12428	0.62542	89

C. Per-Sequence Results

Errors are concentrated (Table V in the appendix): the six sequences with the largest AVE, all aerial, hold 24.6% of the score. The two largest (sequences 29 and 56, AVE 2.22 and 2.20 m/s, RTE 9.7 and 7.9 m) are the two IMUs of one fast flight of a body-frame rig and the next two (37 and 38, AVE 0.94 and 1.13 m/s) the two IMUs of a second flight of the same rig; these four carry 21.8% of the score, and their served speeds exceed those of every labelled recording of that rig. The 11 body-frame aerial sequences average AVE 0.80 m/s and hold 30% of the score; the 34 world-frame ones average 0.18 m/s. Ground errors are uniform and small.

D. Ablation Studies

Fig. 2 is the direct ablation of the numerical stage on the released weights: context-matched raw predictions score 0.164675, the core estimator (base velocity, gravity and bias estimation with composition) 0.142535 and the complete released procedure 0.125051, better on every platform (paired recording-group bootstrap of the complete-minus-raw change −0.0396, 95% interval [−0.0553, −0.0273], 62 of 71 recordings). The exact decomposition $\|\hat{\mathbf{v}} - \mathbf{v}\|^2 = (\|\hat{\mathbf{v}}\| - \|\mathbf{v}\|)^2 + 2(\|\hat{\mathbf{v}}\|\|\mathbf{v}\| - \hat{\mathbf{v}}^\top \mathbf{v})$ separates a speed term from a magnitude-weighted directional term: the complete method reduces vector MSE from 0.065710 to 0.034371 (m/s)², a 47.7% reduction of which 46.9% is speed and 53.1% direction, both components improve on every platform, and platform-macro AVE falls from 0.1301 to 0.0950 m/s (bootstrap interval for the AVE change [−0.051, −0.023]). On the two sensor recordings of the rig’s body-frame validation flight, mean AVE falls from 0.59 to 0.32 m/s.

E. Generalization Analysis

Structure beyond the platform labels. Fig. 3 (appendix) summarises an unsupervised analysis of 59 raw-IMU descriptors of the 475 released recordings. Within the 337 aerial recordings a three-cluster Ward cut agrees with three interpreted sensor groups (adjusted Rand index 0.972): 284 recordings with world-frame accelerometer and gyroscope (bias- and rotation-fitted accelerometer residual 0.113 m/s² under the world hypothesis against 1.838 m/s² under the body hypothesis, lower in 282 of 284), 29 body-frame

¹<https://www.kaggle.com/competitions/tartan-imu-challenge-iro2026/overview/evaluation>

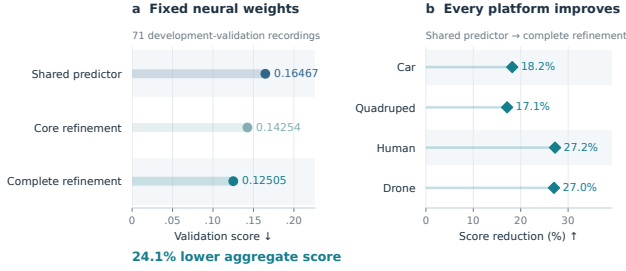


Fig. 2. Fixed-network ablation on the filtered validation set (71 recordings): context-matched neural predictions 0.164675, core estimator 0.142535, complete method 0.125051, a 24.1% reduction with every platform mean improving. The complete method is the released inference procedure. The validation set was used during development, so this is not an estimate of unseen-robot generalisation.

recordings with finer accelerometer quantization and 24 with coarse quantization. Car and quadraped recordings remain mixed in low-order cuts. The numerical branches are hand-designed for these measured conventions, not for the cluster assignments.

Transfer from validation to test. The released method scores 0.1251 on the filtered validation set, 0.1602 on the official service and 0.2059 on the public leaderboard. The gap is aerial: a classifier on raw-IMU descriptors separates validation from test aerial recordings with AUC 0.87 but ground recordings with AUC at most 0.41, and the fast, banked world-frame sub-cluster holds 21% of validation but 55% of test aerial recordings. Validation is therefore a development instrument rather than an estimate of test behaviour for the aerial platform.

F. Failure-Mode Analysis

The dominant failure is the fast body-frame rig: its two test flights reach a served speed of 18.5 m/s where no labelled recording of the rig exceeds 11.3 m/s; on the rig’s labelled flights the raw prediction of an earlier reference network saturates near 8 m/s with a direction error above 10° , and the drag calibration acts on the speed, not on the direction. Second, the world-frame fit is rejected for 4 of 34 world-frame test recordings, which keep the raw prediction. Third, the released frame gate is miscalibrated: its centre misclassifies 95 of the 475 labelled recordings, and recalibrating it at inference costs +0.0018, so the fix belongs in training. Fourth, the human ATE₂₀ floor: within-window rotation makes the one-second mean body velocity an imperfect description of the trajectory that the metric integrates with ground-truth orientation. Unresolved are the aerial branches’ rig-specific constants, the offline design and the absence of a measured unseen-rig evaluation.

V. INSIGHTS AND LESSONS LEARNED

A universal inertial model is feasible for the regression, not yet for the estimation. One shared network with the embodiment inferred internally predicts velocity and speed on all four platforms better than descriptor baselines, and its soft experts never became platform dispatch. What stayed

platform-specific is the physics that turns predictions into a trajectory-consistent estimate: the frame the sensor reports in, the nuisance parameters (biases, scale, lever arm, drag) that explain a recording, and how far to trust the network. One estimator that adapts these under a common formulation is the most important next step, and it may need more data than this benchmark offers.

The bottleneck is coverage and protocol, not capacity.

Extended training, added descriptors and latent readouts gave little, while a few flights outside the labelled speed range dominate the test error. The data with the highest marginal value would have been independent aerial flights with attitude and speed diversity and, on every platform, more independent recording groups rather than more windows. The transferable findings are physical: gravity-aligned conditioning, uncertainty-weighted smoothing with exact mean factors and label-free detection of coordinate conventions do not depend on this benchmark’s platforms; rig-specific calibrations do. Learning-based inertial odometry still lacks a principled treatment of the constant-velocity ambiguity, of convention shifts between sensors and of correlated within-recording errors; differentiable estimation on out-of-fold predictions, group-held-out evaluation and explicit convention detection are the directions we find most promising.

A. Feedback to the Organizers

Two undocumented properties of the data cost the most time. (1) Coordinate conventions: 284 of the 337 labelled aerial recordings report specific force and angular rate in world coordinates with a sign transformation on the gyroscope, the largest source of aerial error for a body-frame model. (2) Data hygiene: nine validation recordings overlap training recordings or carry a mispaired quadraped label, and two training recordings are mispaired. (3) The metric: equal platform weighting puts 57% of the score in 45 aerial sequences of 19 to 60 s that hold 7% of the windows, so the benchmark is an aerial benchmark with a ground sanity check.

VI. CONCLUSION

A single shared-weight Conformer with dual-frame inputs, registers and soft experts predicts body velocity across four platforms, and a parameter-free estimation stage that reconciles its predictions with the recording’s own inertial kinematics reduces the validation score by about a quarter at fixed weights. Improving the network further gave diminishing returns, while a few aerial flights outside the labelled distribution dominate the official error. The most promising continuation is one physical estimator that adapts sensor and rig nuisance parameters under a common formulation, evaluated on rigs held out entirely.

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APPENDIX

A. Sensor Structure Beyond the Platform Labels

a Five clusters describe six populations

475 recordings · Ward clustering in 59 dimensions

Car	56	*	*	*	*
Quadruped	49	*	*	*	*
Human	*	33	*	*	*
Drone: world frame	5	*	279	*	*
Drone: body, finer	*	*	1	21	7
Drone: body, coarse	*	*	*	*	24
	C1	C2	C3	C4	C5

Car and quadruped share a cluster.

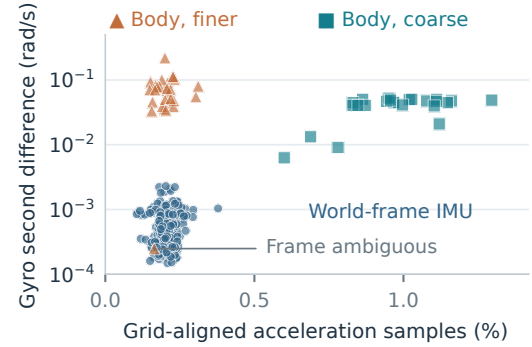
Aerial-only Ward cut: ARI = 0.972

Same assignments with 53 generic descriptors.

Cells: recording counts. Shading: fraction of each row.

Sensor colours: world-frame IMU; body-frame IMU with finer or coarse quantization. No recording is excluded.

b Aerial sensor conventions



c World-frame motion subgroups

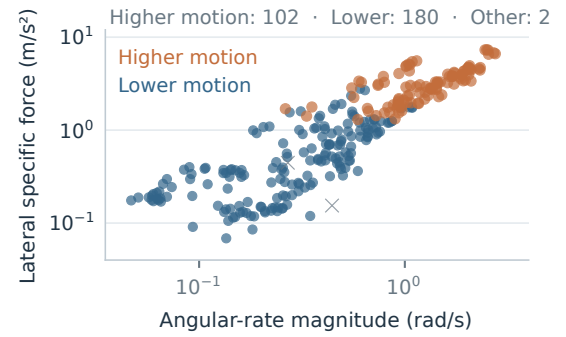


Fig. 3. (Appendix figure referenced in Sec. IV-E.) Sensor and motion structure beyond the platform labels, from raw-IMU descriptors of 475 released recordings. (a) Composition of a five-cluster Ward cut; car and quadruped share a cluster. (b) Two interpretable statistics for the 337 aerial recordings: proximity of accelerometer values to a fixed quantization grid, and the median absolute second difference of the gyroscope. (c) World-frame recordings split by motion statistics. Clustering uses IMU descriptors only; the interpretation uses the released attitude and velocity labels.

B. Staged Training Curves

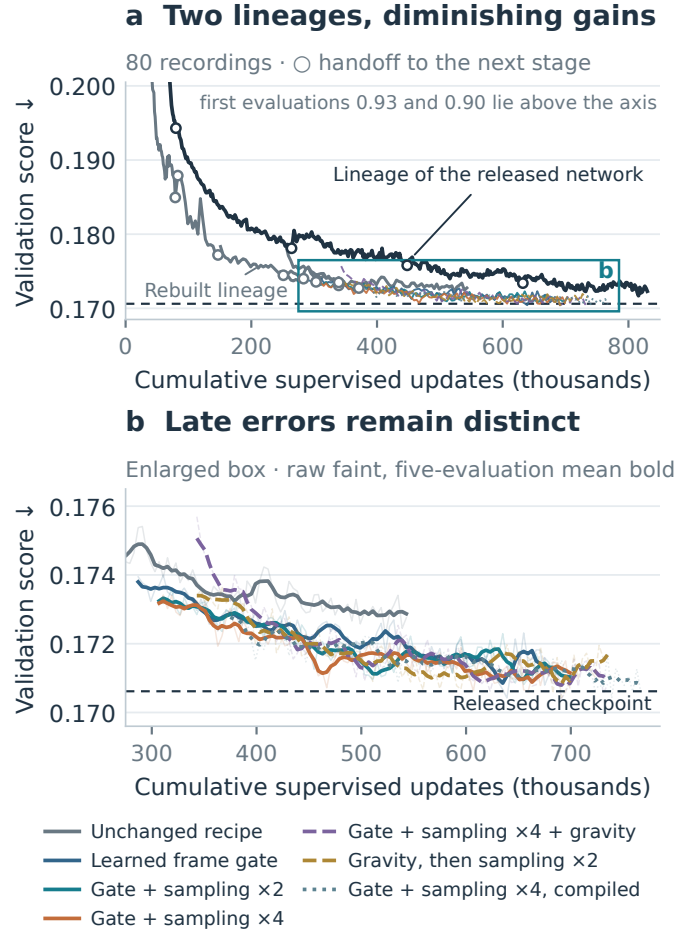


Fig. 4. (Appendix figure referenced in Sec. III.) Staged training on the full validation set (80 recordings) against cumulative supervised updates along the inherited checkpoint lineage: a stage’s abscissa adds the updates of the parent checkpoint it loaded, and the shared 15,000 pretraining updates are excluded. (a) The two supervised lineages over their whole recorded history: the lineage of the released network (dark) and a second lineage rebuilt from the same pretraining with descriptor-assisted formation and an earlier soft-expert stage (grey), which served only as the trunk for the seven extended recipe continuations (colours) and did not produce the released network. Circles mark checkpoint handoffs, where a stage’s validation-minimum checkpoint became the next stage’s start; the box outlines panel (b). (b) The seven continuation recipes, raw evaluations faint and a centred five-evaluation mean bold; sampling is the aerial sampling weight, gravity the gyro-aided gravity conditioning, compiled the CUDA-graph execution. The released checkpoint is the dashed reference in both panels because an intervening parent step is unresolved. Distinguishable late curves are descriptive evidence of diminishing returns, not a statistical equivalence result.

C. Per-Sequence Results (Full Table)

TABLE V

PER-SEQUENCE RESULTS OVER THE 89 TEST SEQUENCES FROM THE OFFICIAL PER-SEQUENCE SCORING SERVICE (SUBMISSION MD5 6d371534ca010cdc7fdc8f18d00fc6a0). ATE₂₀ AND RTE IN M, AVE IN M/S. **RTE** (RELATIVE TRAJECTORY ERROR) FOLLOWS STURM *et al.* [2]: OVER SLIDING WINDOWS OF $\Delta t = 5$ s IT IS THE RMSE OF THE DIFFERENCE BETWEEN PREDICTED AND GROUND-TRUTH RELATIVE DISPLACEMENT; IT MEASURES LOCAL DRIFT AND DOES NOT ENTER THE RANKING. SEQUENCE NUMBERS ARE THE SERVICE’S: THE 45 AERIAL RECORDINGS LIE WITHIN 1–88, THE 16 QUADRUPEL WITHIN 6–85, THE 10 HUMAN WITHIN 8–86 AND THE 18 CAR RECORDINGS WITHIN 10–89.

Seq.	ATE ₂₀ ↓	AVE ↓	RTE ↓	Seq.	ATE ₂₀ ↓	AVE ↓	RTE ↓	Seq.	ATE ₂₀ ↓	AVE ↓	RTE ↓	Seq.	ATE ₂₀ ↓	AVE ↓	RTE ↓
1	0.81318	0.14567	1.82103	24	0.46518	0.02730	0.17117	47	1.05252	0.39140	1.55138	70	0.37672	0.05293	0.24538
2	0.49378	0.20999	1.03591	25	0.11033	0.03267	0.10865	48	0.22909	0.01011	0.07301	71	0.16135	0.04609	0.14693
3	0.37798	0.16852	0.56161	26	0.45772	0.04448	0.25675	49	0.79890	0.14162	1.41995	72	0.65138	0.29255	1.09211
4	0.31160	0.14769	0.53211	27	0.54208	0.03611	0.24837	50	0.27090	0.13830	0.50211	73	0.40440	0.13357	0.34905
5	0.41945	0.16241	0.74109	28	0.78294	0.23062	1.13309	51	0.10433	0.07424	0.14285	74	0.18100	0.10890	0.37528
6	0.68983	0.09005	0.74610	29	1.55962	2.21741	9.72192	52	0.13405	0.07532	0.18146	75	0.53658	0.23846	1.09168
7	0.67710	0.13920	0.68556	30	0.68726	0.33993	1.38601	53	0.41201	0.27295	0.52407	76	0.10921	0.03627	0.09872
8	0.32099	0.02328	0.15357	31	0.64141	0.13835	1.56093	54	0.16288	0.04439	0.16892	77	1.12065	0.24054	2.31102
9	0.20800	0.07242	0.29284	32	0.18990	0.06415	0.16432	55	0.37214	0.18127	0.84888	78	0.12380	0.02291	0.08059
10	0.81682	0.08034	0.68976	33	0.13928	0.04101	0.12690	56	2.04422	2.19920	7.89591	79	0.24568	0.16518	0.65191
11	1.01793	0.30694	2.13002	34	0.42266	0.14033	0.69865	57	0.85165	0.15845	2.04626	80	0.42749	0.21511	1.01852
12	0.38251	0.16791	0.83032	35	0.99680	0.18820	1.05054	58	0.22924	0.04698	0.24795	81	0.30055	0.05450	0.21020
13	0.15005	0.03640	0.15270	36	0.95368	0.15174	1.79630	59	0.23207	0.06045	0.21704	82	1.57956	0.24313	1.61520
14	0.52451	0.22219	1.06459	37	0.78572	0.93562	4.40932	60	0.14698	0.07558	0.19232	83	0.44033	0.07208	0.35699
15	1.20874	0.06391	0.42432	38	0.82869	1.12899	5.37702	61	0.82377	0.22611	1.15669	84	0.49172	0.03953	0.29533
16	0.13054	0.03655	0.20843	39	0.23083	0.07553	0.36913	62	0.12162	0.01849	0.05547	85	0.09113	0.03893	0.12344
17	0.66036	0.30787	1.43130	40	0.76454	0.21535	1.58361	63	0.35314	0.04161	0.26006	86	0.26203	0.01581	0.10713
18	1.50415	0.61243	2.70248	41	1.00675	0.25467	2.09935	64	0.49474	0.03959	0.26758	87	0.28334	0.14880	0.61727
19	0.49495	0.20465	0.93061	42	0.31595	0.09739	0.25023	65	0.07739	0.04170	0.10083	88	1.00769	0.20285	2.05804
20	0.16044	0.05381	0.35263	43	0.56369	0.12373	0.61214	66	2.07920	0.23080	1.61605	89	0.10242	0.02242	0.07267
21	0.80665	0.14641	1.29782	44	0.23453	0.05213	0.31884	67	0.13063	0.03295	0.15916				
22	0.52013	0.22585	1.02863	45	0.14932	0.06319	0.16201	68	0.51182	0.04630	0.20445				
23	1.03605	0.17591	2.05679	46	0.26596	0.04051	0.18756	69	0.58906	0.05247	0.35995				

D. Reproducibility and Submission Artifacts

Submission. Kaggle team *Marco*; Kaggle submission ID 56407636, selected for final scoring; submission CSV MD5 6d371534ca010cdc7fdc8f18d00fc6a0 (30,644 rows plus header), public leaderboard score 0.20589.

Weights and code. The released package contains `inference.py`, `model.pt` (SHA-256 184f6e1a66019e9180f6cf7a7b51aa46976d08c3f11785c0a0026690ef84b17b; 388 float32 tensors, 41.3 M values), pinned requirements (Python 3.10, PyTorch 2.5.1, NumPy 2.2.6, pandas 2.3.3, SciPy 1.15.3), the submitted CSV and a README. Running the script on the raw test recordings and window index reproduced the submitted CSV byte for byte (2026-09-24; Apple M5 Pro; 515.4 s). This rerun used OS-level network denial and a 16 GB MPS allocator cap; sampled peak driver allocation was 1.43 GiB. File-access auditing found no stored prediction inputs. The reported score is a single deterministic inference of one training run; no seed averaging is involved. Exact CSV bytes were verified on Apple MPS with a fixed five-thread CPU pass; native CUDA execution has not been measured.

Compliance declaration.

- 1) **Single model, shared weights.** Yes. All predictions come from one network with one shared set of weights. The trunk and velocity heads use a validation-selected exponential moving average checkpoint from one training lineage; the uncertainty and platform-classification heads were fitted afterwards on the frozen trunk using training and validation recordings. The uncertainty head consumes pooled window features and the platform head consumes mean register features; they supply numerical-estimator weights and routing, respectively. The three internal expert mixtures are dense, token-level softmax mixtures inside the shared trunk; every expert is evaluated for every token on every platform. There are no separately trained per-platform networks.
- 2) **Platform classification and routing.** Yes, after the network. The network’s platform posterior (mean register embedding, four logits, averaged over tiles) selects which settings of the parameter-free numerical estimator apply to a recording and whether the aerial branches run: recordings with ground mass at least 0.9 use the smoother settings of their most likely ground platform; otherwise the aerial branch is used when the drone mass is at least 0.5 or the raw 90th-percentile speed exceeds 5.86 m/s; unrouted recordings keep the raw prediction. Within the aerial branch a label-free noise-floor statistic of the raw IMU selects the world-frame or body-frame physics and a quantization statistic selects the coarse-quantization settings. No ground truth and no trajectory identifier enters inference.
- 3) **Ensembling.** No model ensemble: one network with one set of weights, and no rotation or other input-perturbation TTA. The only test-time augmentation is overlapping-tile evaluation. A recording longer than the context is tiled at stride 30 windows and each window’s outputs are averaged over every tile that covers it (typically $L/30$ tiles: 2, 4 or 8 at the 60-, 120- and 240-window contexts), with cosine-weighted velocities (taper floor 10^{-3}), uncertainties as variances and platform posteriors as tile means; a recording that fits in the context, as every aerial recording does, is one tile with no averaging. The same weights run in four such sweeps (120 windows on the accelerator for base estimation and the world-frame fit, 120 windows on the CPU for the refinement inputs and the raw-direction hedge, 60 windows for body-frame aerial refinement, 240 windows for humans). The sweeps are not averaged as an ensemble; each feeds a stage of the deterministic numerical estimation of Sec. II-E. The deployed weights are one EMA checkpoint (decay 0.999) of one training run; no soup or averaging of separately trained checkpoints.
- 4) **Test-set leakage.** The encoder was pretrained on unlabelled training and validation IMU; supervised velocity training used the training split only. The uncertainty and platform heads were fitted on training and validation recordings, so validation is not held

out from the full system. No external data, external pretrained weights or test recordings entered pretraining or training. No test label or trajectory identifier enters inference. Within-stage checkpoints were selected on validation. The numerical stage has no learned parameters: its smoother settings were fitted out-of-fold or on validation recordings, and its fixed calibration constants are applied only inside regimes read from the recording’s own statistics (Sec. II-E).

- 5) **Development versus submitted system.** The submitted predictions are the output of the released procedure exactly: the released source is a tidied rewrite of the development code and reproduces the submitted CSV byte for byte (appendix, Reproducibility and Submission Artifacts).

E. Final Predictions

The submitted CSV (`window_id, vx, vy, vz`, 30,644 rows plus header, MD5 `6d371534ca010cdc7fdc8f18d00fc6a0`) is included in the released package as `submission.csv` and is the file scored on Kaggle and by the per-sequence service. The numerical estimation stage of the Method post-processes the overlapping-tile predictions; TTA is described in compliance question 3. Every window of every test trajectory is predicted; no degenerate windows occur.

F. Model and Inference Code

Weights, inference script, pinned requirements and the submitted CSV: <https://huggingface.co/Marco-intelligence/tartanimu-marco>. Reproduce with `python inference.py --test_root <npz dir> --windows test_windows.csv --out predictions.csv`. Training code is available upon request. Sec. II-D and the architecture literal in `inference.py` document the network, and the numerical stage is fully specified by the released source.